

Battery Datasheet

FLPB301031

2021. 02. 16.

Routejade R&D Center



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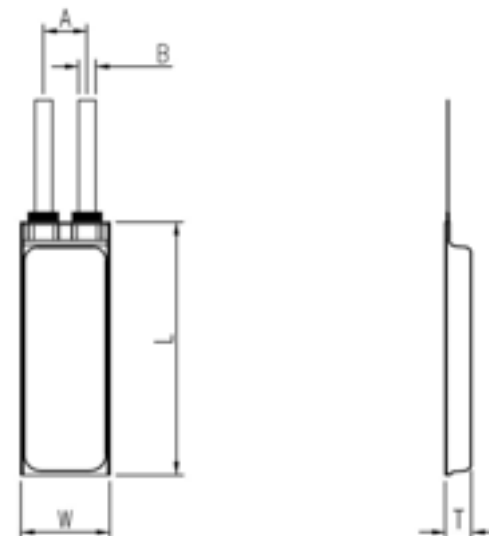
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I. GENERAL SPECIFICATION

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1. General Specification

Item		Specification	Remark
1	Nominal Capacity	76mAh	0.2C rate, 3.0V cut-off
2	Minimum Capacity	70mAh	0.2C rate, 3.0V cut-off
3	Nominal Voltage	3.8V	From 4.35V to 3.00V
4	Charging Method	CC/CV	Constant Current / Constant Voltage
5	Charging Current (Std.)	0.5C (35.0mA)	0 ~ 45°C
6	Charging Current (Max.)	1.0C (70.0mA)	
7	Charging Voltage	4.35 ± 0.03V	
8	Charging End Condition	0.03C (2.1mA)	At CV mode
9	Charging Time (Std.)	< 3.0 hours	
10	Discharge Current (Std.)	0.2C (14.0mA)	- 20 ~ 60°C
11	Discharge Current (Max.)	2C (140mA)	
12	Discharge Cut-off Voltage	3.0V	
13	Cycle	>80% @500cycle	0.5C/0.5C cycle @25°C
14	Thickness (mm)	Max. 3.0mm	(@Swelling max.)
	Width (mm)	Max. 10.2mm	
	Length (mm)	Max. 29.1mm	
15. Storage Temperature Range		40 ~ 60°C	For up to one week
		25 ~ 40°C	For up to three month
		-20 ~ 25°C	For up to one year
16	Impedance (mΩ)	< 400	
17	Weight (g)	< 1.8	



	Item	Spec.
T	Thickness	3.0mm max.
W	Width	10.2mm max.
L	Length	29.1mm max.
A	Distance between centers of 2 tabs	5.0 ± 0.5mm
B	Tab width	2.0 ± 0.1mm

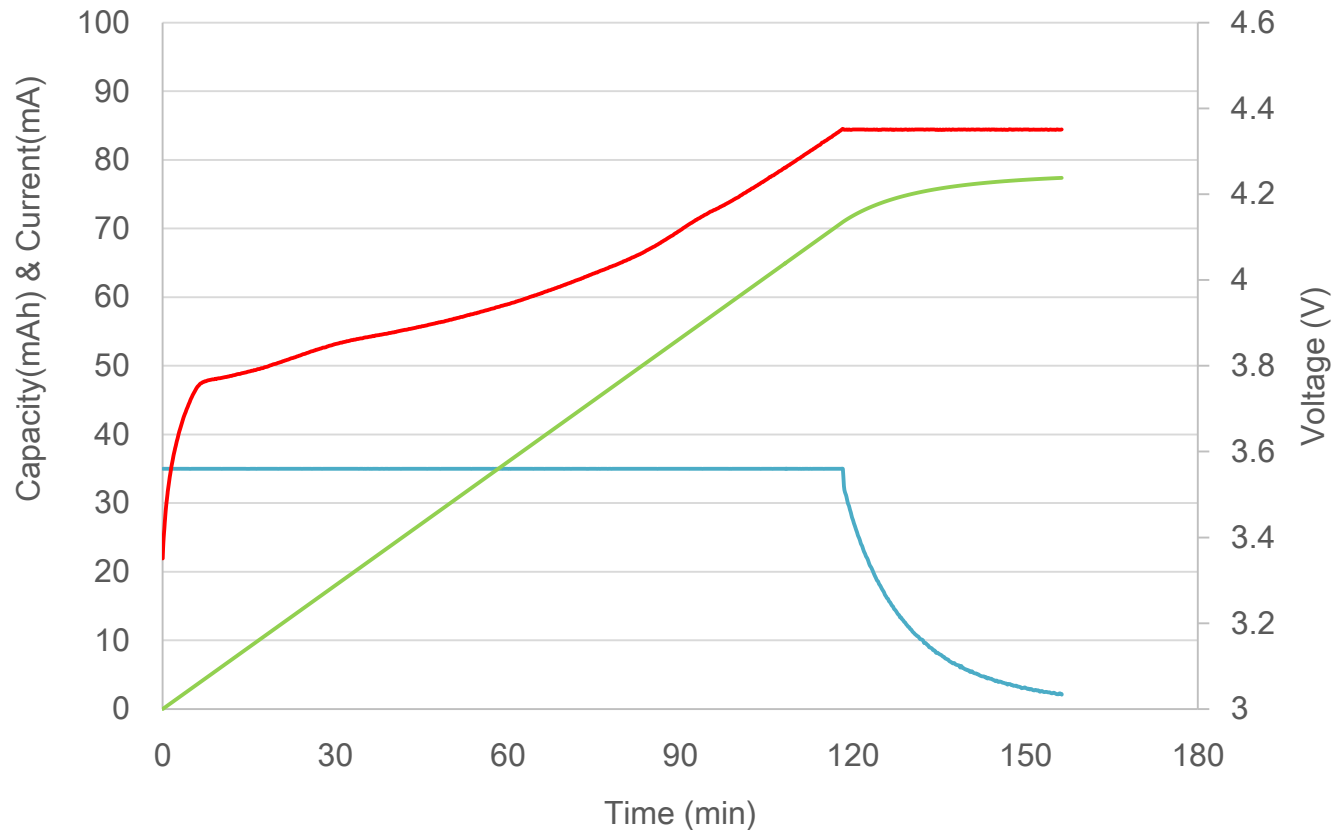
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II. BATTERY CHARACTERISTICS

1. Charge Characteristics
2. Discharge Characteristics
3. Cycle(1.0C/1.0C)
4. Cycle(1.0C/1.0C, 0.5C/0.5C)
5. 60°C, 90%RH Storage Test

1. Charge Characteristics at 25°C

Charge Characteristic at 25°C



Model : FLPB301031

Minimum

Capacity : 70mAh

Charge Condition

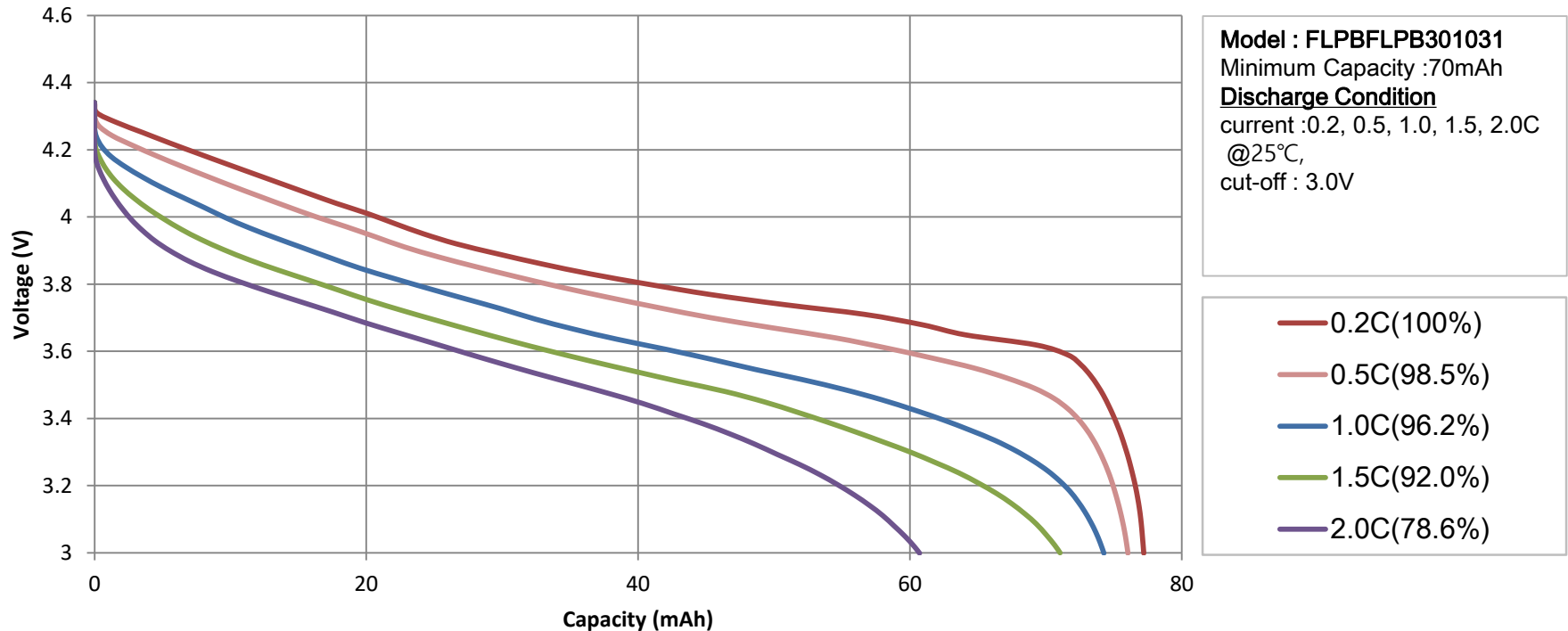
current : 0.5C @25°C

cut-off : 4.35V, 0.03C

— current
— Capacity
— Voltage

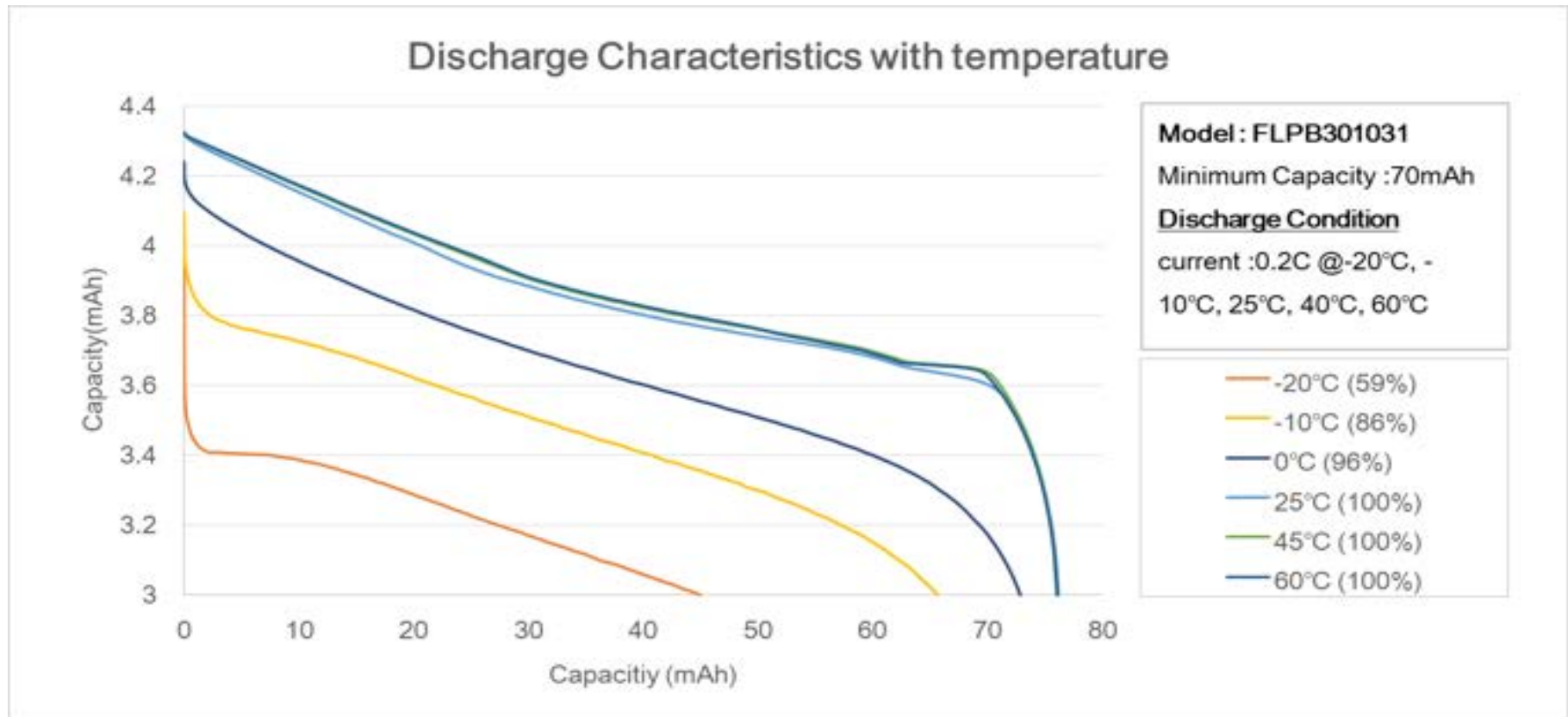
2. Discharge Characteristics at 25°C

Discharge Characteristics at 25°C



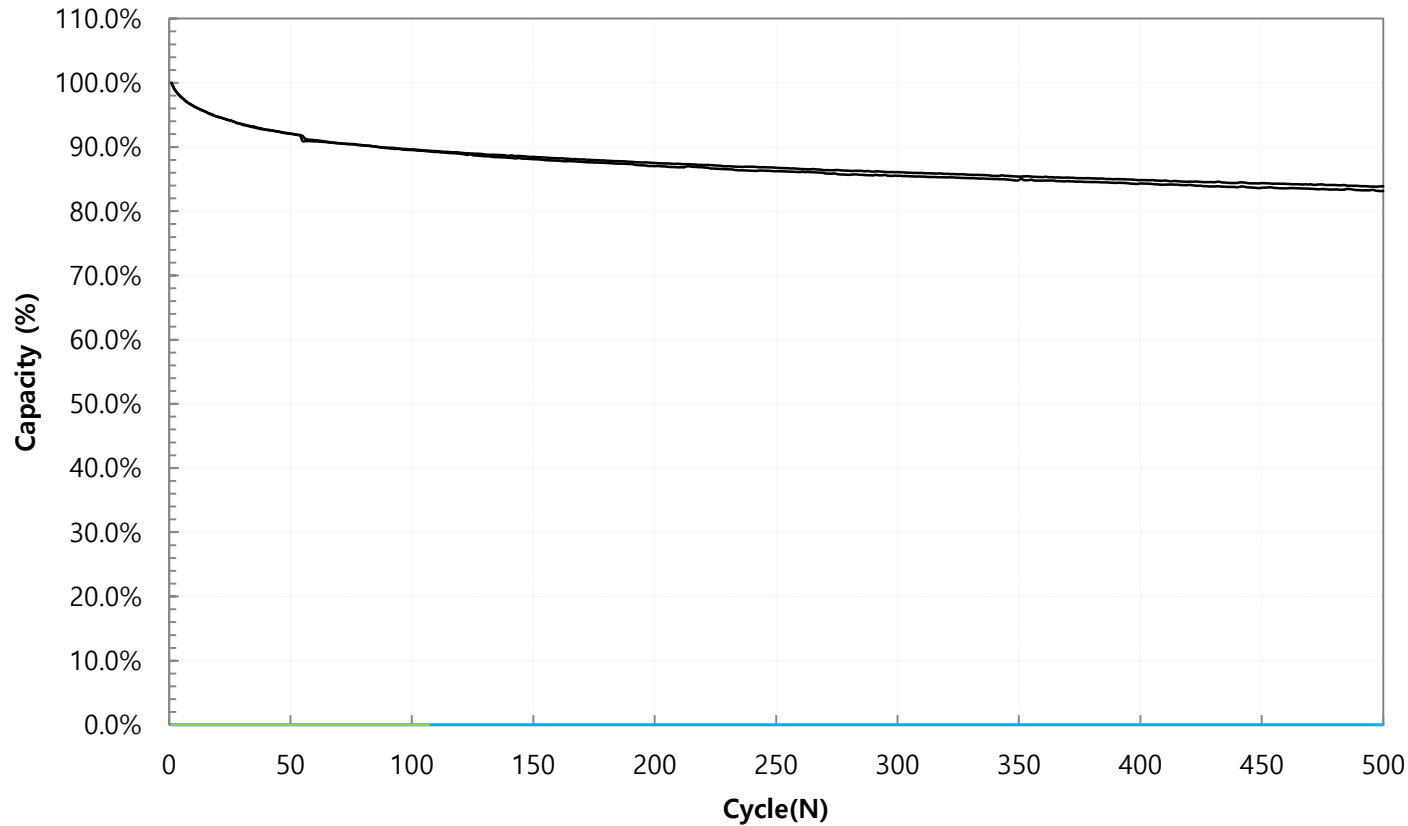
Model	0.2C	0.5C	1.0C	1.5C	2.0C
mAh	77.2	76	74.3	71.1	60.7
%	100%	98.5%	96.2%	92.0%	78.6%

3. Discharge Characteristics with Temperature



Model	-20°C	-10°C	0°C	25°C	40°C	60°C
mAh	45	66	73	76	76	76
%	59%	86%	96%	100%	100%	100%

4-1. Cycle(1.0C/1.0C)



Model : FLPB301031
Minimum Capacity :
70mAh

Charge condition
current : 1.0C @25°C
voltage : 4.35V
cut-off : 0.03C

Discharge Condition
current : 1.0C @25°C
cut-off : 3.0V

— FLPB301031_1

— FLPB301031_2

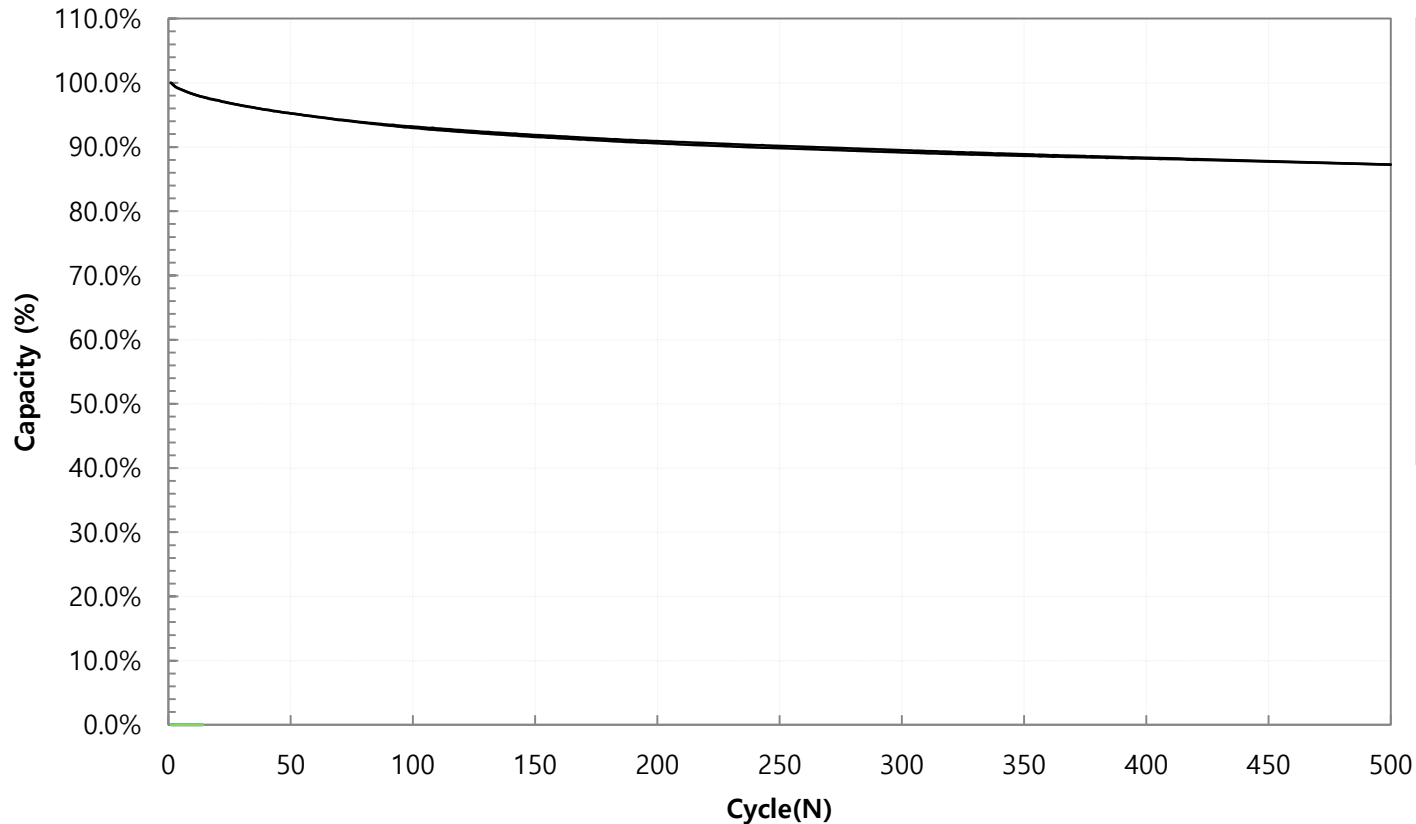
Test Condition

1C/1C at 25°C

%

83.2% @ 500th

4-2. Cycle(0.5C/0.5C)



Model : FLPB301031
Minimum Capacity :
70mAh

Charge condition
current : 0.5C @25°C
voltage : 4.35V
cut-off : 0.03C

Discharge Condition
current : 0.5C @25°C
cut-off : 3.0V

— FLPB301031_1
— FLPB301031_2

Test Condition	0.5C/0.5C at 25°C
%	87.3% @ 500th

5. 60°C, 90%RH Storage Test

** Average of 5 samples.*

Category	Before storage	After storage	Δ
Weight(g)	1.56	1.56	0%
Thickness, mm (at RT)	2.762	2.766	0.1%
Voltage(Vdc)	4.3339	4.2301	- 2.4%
Impedance(m Ω)	175	284	62%
Residual capacity (mAh)	76.10	61.81	81.2%
Recovery capacity (mAh)	76.10	69.10	90.8%

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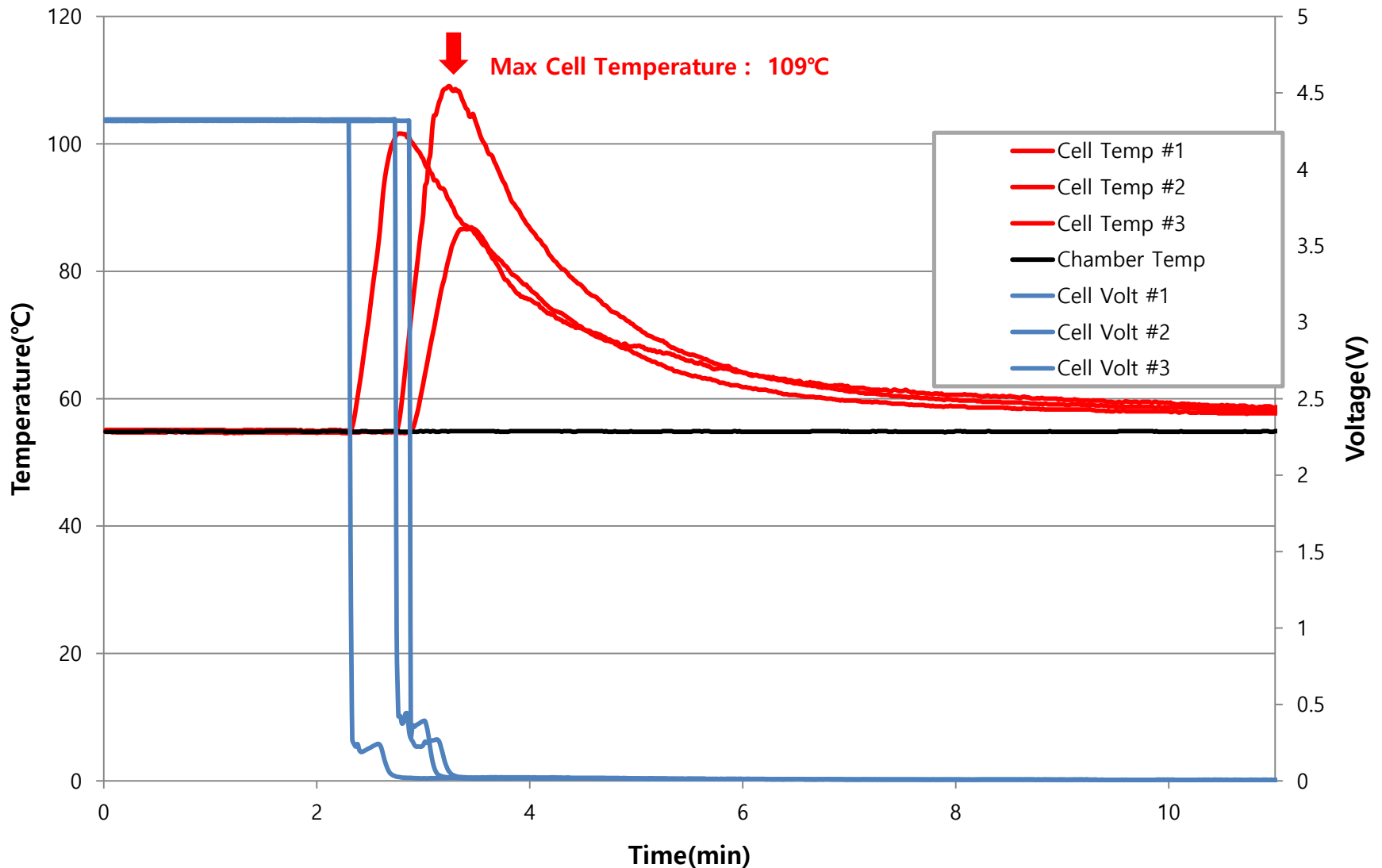
III. BATTERY SAFETY TEST

1. Charge Characteristics
2. Discharge Characteristics
3. Cycle(1.0C/1.0C)
4. Cycle(0.5C/0.5C)
5. 60°C, 90%RH Storage Test

III. Safety Test

No.	Test Name	Requirements(Criteria)	Result
1	55°C External Short Circuit (UL 1642)	No fire, No explosion, No venting. or cell temperature should not exceed 150°C.	Pass (NF, NE, NS) (Max Temp : 109.0°C)
2	Impact Test	No fire, No explosion.	Pass (NF, NE, NS) (Max Temp : 113.7°C)
3	Nail Penetration Test (RS C 0017)	No fire, No explosion No venting.	Pass (NF, NE, NS) (Max Temp : 86.3°C)
4	150°C Heating Test (5°C/min rate, 10min)	After cell temperature reaching at 150±2°C, No fire, No explosion No venting	Pass (NF, NE, NS) (Max Temp : 148.5°C)
5	Overcharge Test (5V, 3C)	No fire, No explosion No venting or cell temperature should not exceed 150°C.	Pass (NF, NE, NS) (Max Temp : 47.2°C)

1. 55°C External Short Circuit Test



1. 55°C External Short Circuit Test

Before testing

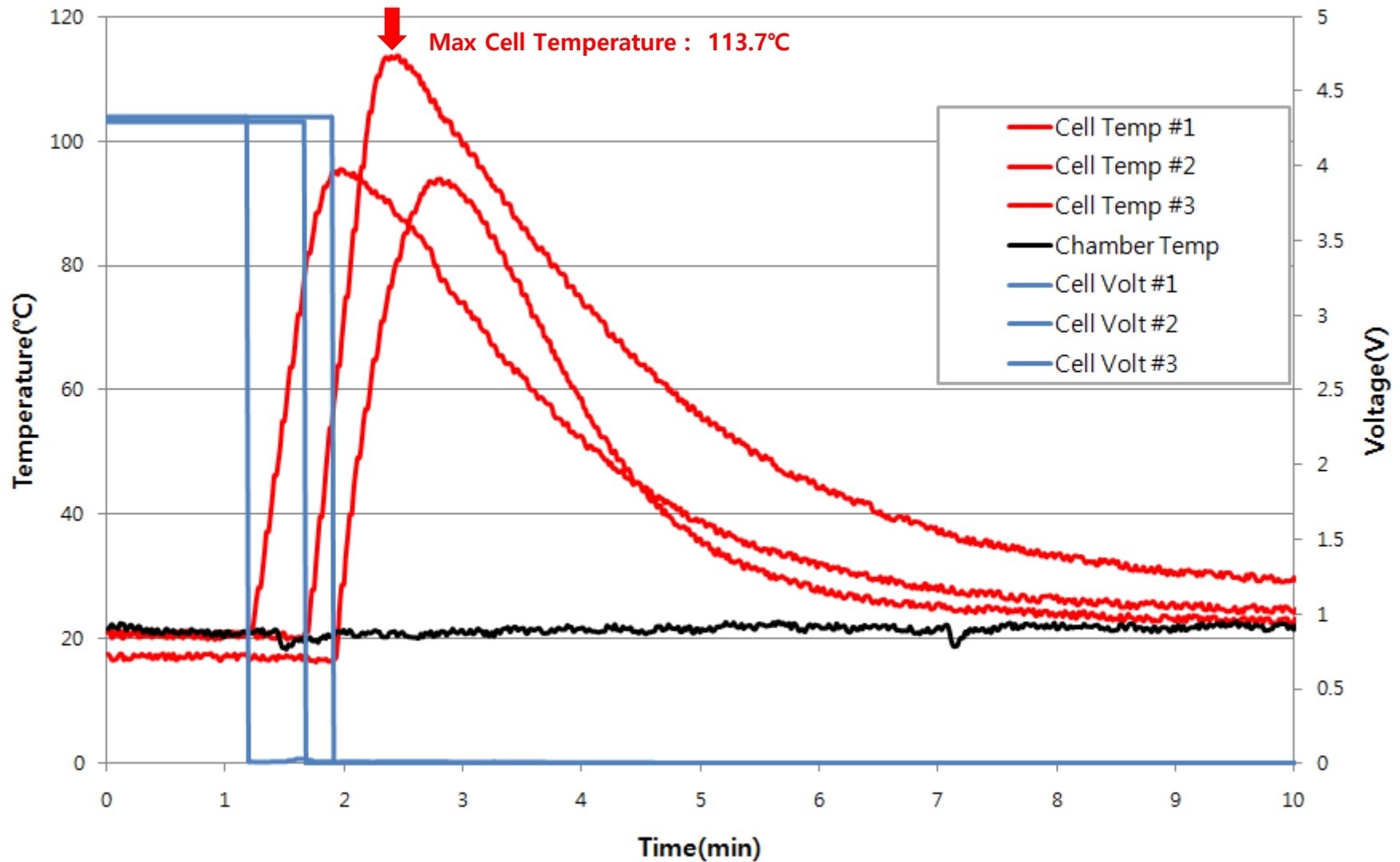


After testing



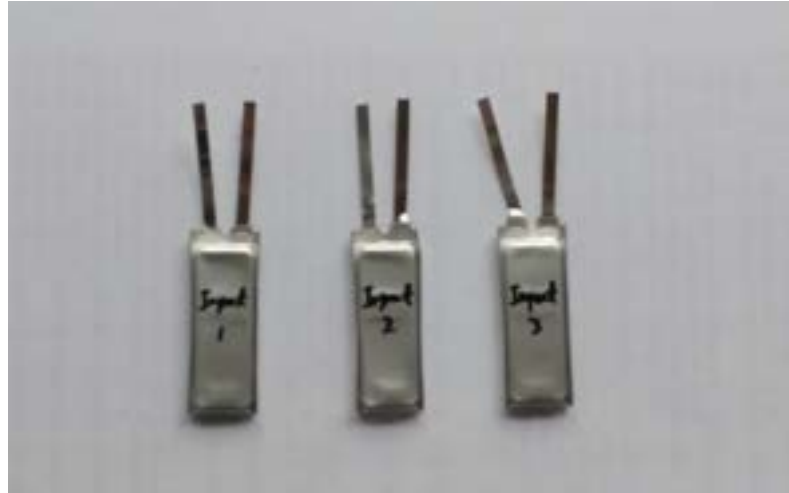
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2. Impact Test



2. Impact Test

Before testing

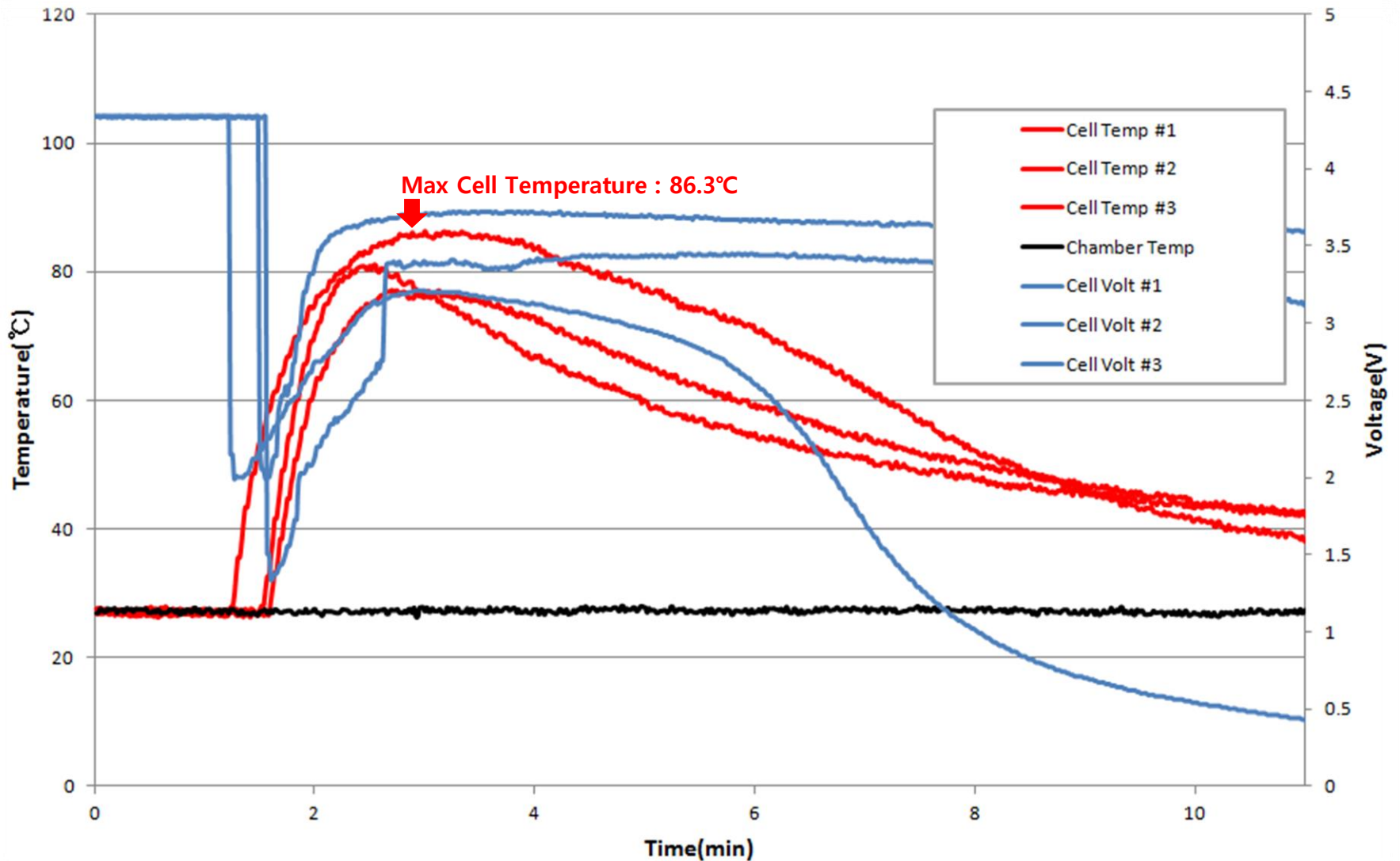


After testing



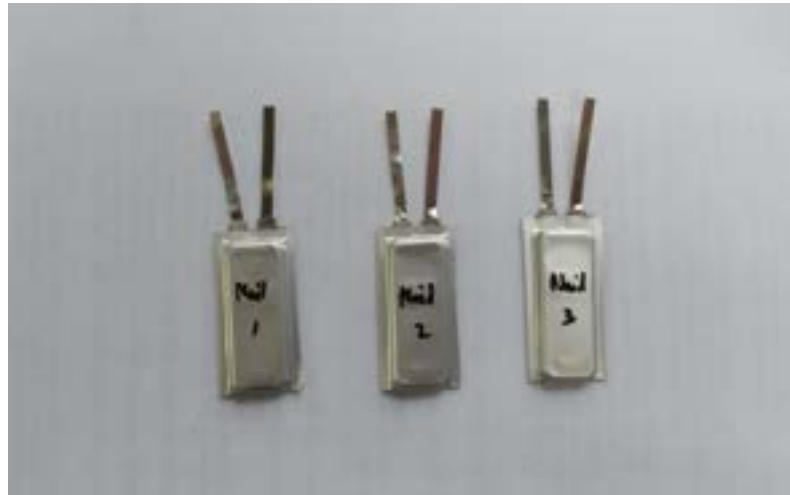
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3. Nail Test



3. Nail Test

Before testing

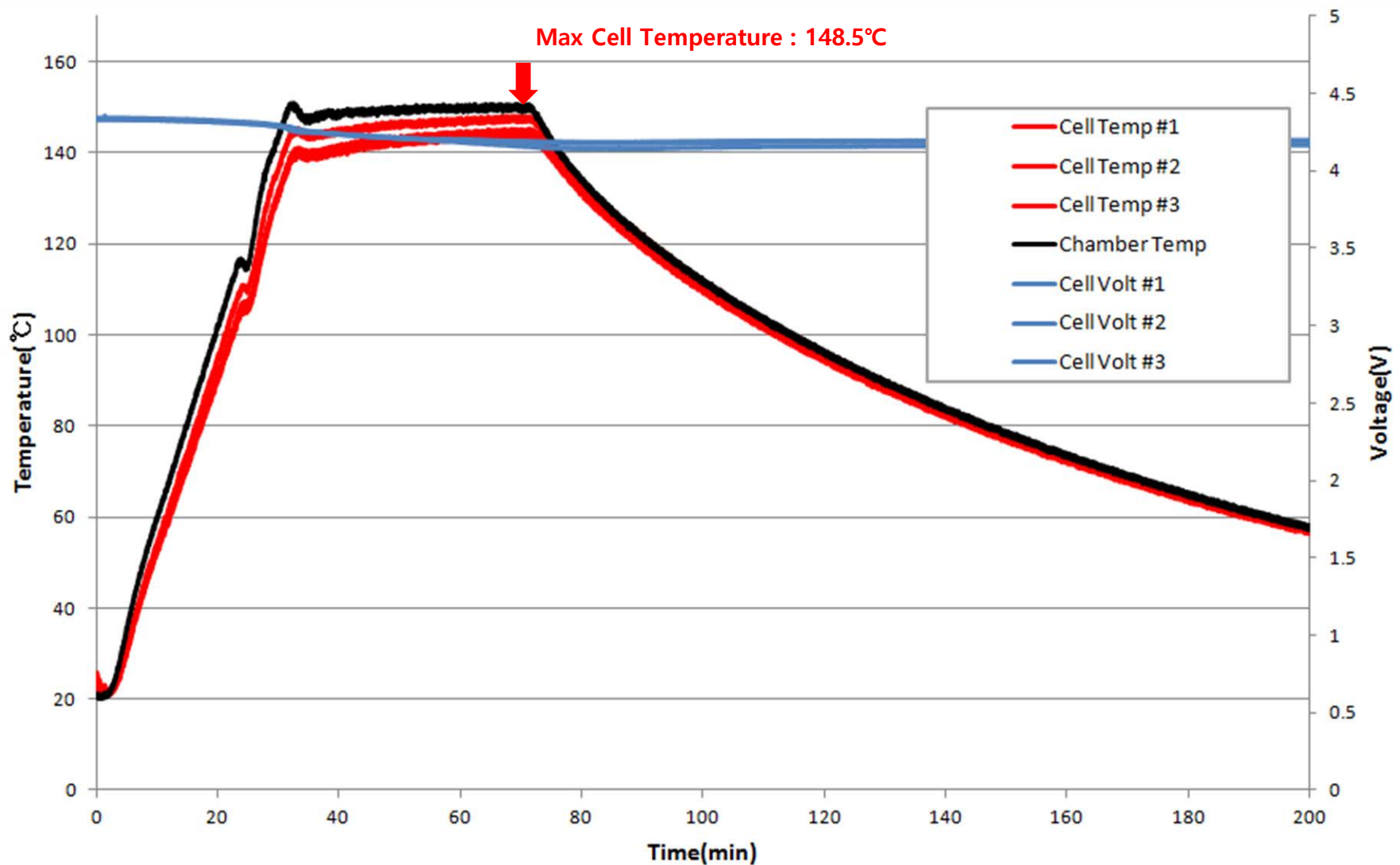


After testing



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4. 150°C Heating Test



4. 150°C Heating Test

Before testing

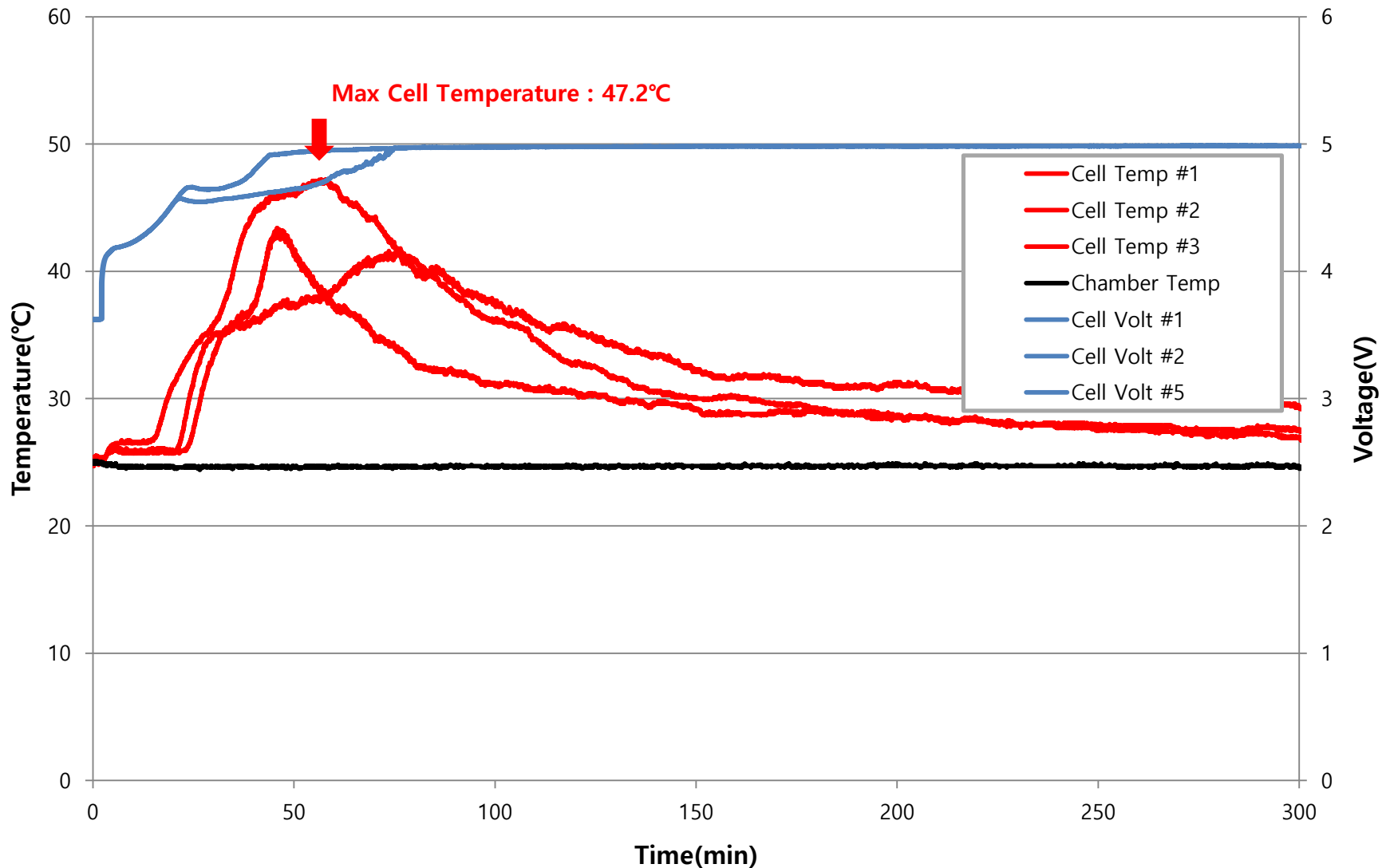


After testing



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5. 5V,3C Overcharge Test



5. 5V,3C Overcharge Test

Before testing



After testing



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